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A Pipeline for Top-Down Spatial Biodiversity Modeling: Development, Evaluation, and Future Extensions

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Contents

1 Introduction

1.1 Motivation

Biodiversity is defined as "the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."[1] In this sense, biodiversity refers to the biotic component of ecological systems: the variety and variability of living organisms, from plants and animals to fungi, bacteria, and protists. Biodiversity is integral to ecosystem functioning and stability, supporting processes such as nutrient cycling, pollutant breakdown, water regulation, carbon storage, and ecosystem productivity (Stork 2009; Cardinale et al. 2012; Gascon et al. 2015). Furthermore, biodiversity is fundamental to human survival and well-being. It supports food security through pollination, soil fertility, fisheries, and livestock systems (Gascon et al. 2015; Brauman et al. 2020). It also provides biological materials and natural compounds used in medicine and drug discovery, including antibiotics such as penicillin (Gascon et al. 2015). Beyond these material contributions, biodiversity supports regulating ecosystem services such as water purification, climate regulation, and carbon storage, while also contributing to biomass-based energy systems and sustaining human health, livelihoods, and well-being (Brauman et al. 2020; Naeem et al. 2016; Kim et al. 2024).

Despite the overwhelming benefits of biodiversity to humanity and ecosystems, biodiversity is declining at an unprecedented rate. Approximately 74% of the world's terrestrial wilderness has experienced degradation, and approximately 9.6% has been completely lost since the early 1990s (Watson et al. 2016). Marine ecosystems have also been heavily affected, with 66% of the ocean's surface having been significantly impacted by human activities (Díaz et al. 2019). Moreover, it is reasonably likely that most coral reefs will be lost if global temperatures exceed 2°C (Frieler et al. 2012). Furthermore, human activities threaten the survival of approximately 1 million species, with approximately 25% of plants and animals threatened (Díaz et al. 2019). Species extinction rates have also increased and are now estimated to be around 1,000 times higher than the natural background rate (Pimm et al. 2014). Compounding this challenge, conservation efforts are underfunded, with a deficit exceeding half a trillion US dollars annually (Deutz et al. 2020). The acceleration of biodiversity loss and the drastic underfunding of conservation efforts have necessitated a shift towards the prioritization of biodiversity conservation and ecological rehabilitation.

Many initiatives around the world are being adopted to combat biodiversity loss and climate change, which are intricately linked. These initiatives focus on international cooperation, expanding protected areas, reducing pollution, and providing incentives for sustainable practices (IPBES 2019; IPCC 2023). Specific policy initiatives such as the 30x30 target—which aims to protect 30% of the world's land area by 2030—represent critical efforts for ecological recovery. If successful, they could generate large gains and rebounds in biodiversity. Achieving these goals would have significant impacts, including carbon sequestration, improved water quality regulation, and expanded capabilities to mitigate nutrient pollution (Zeng, Koh, and Wilcove 2022). For these initiatives to be successful, it is crucial to have accurate and robust models that can predict biodiversity patterns and trends. These models can inform conservation strategies, identify priority areas for protection, and assess the potential impacts of different conservation actions (Sinclair, White, and Newell 2010).

In this thesis, we use the broad term *Spatial Biodiversity Models* to refer to models that predict one or more biodiversity measurements across space. This terminology emphasizes the spatially explicit nature of these models, which relate biodiversity patterns to environmental conditions at particular locations (Ferrier and Guisan 2006). Moreover, models that predict community-level biodiversity metrics are referred as community-level biodiversity models or community-level models in the literature (Ferrier, Drielsma, et al. 2002; Nieto-Lugilde et al. 2018). These community-level biodiversity metrics can include species richness, total abundance or relative abundance, intactness, or compositional similarity (Lamb et al. 2009; Santini et al. 2017). Broadly, community-level modeling approaches can be grouped into three main strategies based on how they produce predictions for community-level biodiversity metrics (Ferrier and Guisan 2006): (1) Bottom-up ("predict first, assemble later") approaches, which commonly use species distribution models (SDMs) to predict the spatial distribution of individual species based

[1] Convention on Biological Diversity (1993) *Article 2. Use of terms.* Available at: https://www.cbd.int/convention/articles/?a=cbd-02 (Accessed: 8 April 2026).

on environmental variables; multiple SDMs can be aggregated or stacked to estimate community-level metrics (Ferrier and Guisan 2006; Distler et al. 2015; Beery et al. 2021); (2) "assemble and predict together" approaches, which frequently use joint species distribution models (JSDMs) to model multiple species simultaneously, accounting for shared environmental responses and residual species associations (Warton et al. 2015; Wilkinson et al. 2019); and (3) top-down ("assemble first, predict later") approaches, in which the model directly predicts community-level biodiversity metrics rather than first modeling individual species or assemblages (Ferrier and Guisan 2006; Pollock et al. 2020).

While each strategy has its own strengths and weaknesses and excels in different contexts, it is unlikely that one modeling strategy will consistently outperform the others (Ferrier and Guisan 2006; C. Zhang et al. 2019). However, bottom-up approaches, are the most extensively researched modeling strategy for community-level biodiversity prediction. Many studies have focused on improving species distribution models, including benchmark evaluations (Norberg et al. 2019; Valavi et al. 2022), reviews of specific models (Waldock et al. 2022), and guidance on best practices (Araújo et al. 2019). Similarly, "assemble and predict together" models have also been the focus of many studies. Joint species distribution models have been developed partly to address the limited treatment of species co-occurrence and biotic interactions in traditional species distribution models (Wilkinson et al. 2019; Escamilla Molgora et al. 2022). Top-down models, on the other hand, are less widely studied and underutilized (Pollock et al. 2020). A significant gap remains in our understanding, comparison, and evaluation of bottom-up and top-down models, both between these modeling approaches and within the different methods they encompass. This highlights the need for further research and methodological development of top-down models (Ferrier and Guisan 2006; Pollock et al. 2020).

Currently, top-down models are often evaluated in individual studies that use different datasets, model types, biodiversity metrics, and evaluation procedures (Mondal and Bhat 2021; Cai et al. 2023; Ngor et al. 2023; Baggström et al. 2025). This can make it difficult to understand the current landscape of top-down models. While there have been some reviews of community-level modeling compared to other spatial biodiversity models (Nieto-Lugilde et al. 2018), there still remains a lack of best practice guidelines for top-down models, comparable to those available for bottom-up approaches (Araújo et al. 2019). This has prompted the need for a pipeline for models using a top-down strategy to allow for more rigorous testing of these models (Pollock et al. 2020). This pipeline would allow for a more systematic evaluation of different top-down models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a pipeline would enable a more nuanced evaluation of these models and their applicability across different ecological contexts, and allow for quick comparison across different models. This would ultimately contribute to the advancement of biodiversity modeling as a whole and a deeper exploration of the potential of top-down models.

1.2 Research Objectives and Scope

The primary objective of this master's thesis is to develop a pipeline for top-down modeling strategies that predict community-level biodiversity metrics and to provide a case study demonstrating the applicability and flexibility of this pipeline. To achieve this objective, a pipeline is first developed that integrates generalized data preprocessing, feature engineering, model training, and evaluation steps for top-down models. This pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches. Next, a case study is conducted using this pipeline to compare the performance of different models on derived biodiversity metrics from the United Kingdom Butterfly Monitoring Scheme (UKBMS) dataset (UK Butterfly Monitoring Scheme 2026), accessed through the Global Biodiversity Information Facility (GBIF)[2]. Models used in this case study will include statistical models, specifically generalized linear models and generalized additive models (Nelder and Wedderburn 1972; T. J. Hastie and Tibshirani 1986), as well as machine learning models, including random forests, gradient boosting, and artificial neural networks (Breiman 2001a; Friedman 2001; Lek et al. 1996). These results are then analyzed to provide insights into the strengths and weaknesses of different top-down models across different contexts. By applying the pipeline across contexts that vary in spatial scale, environmental predictor sets, and evaluation strategy, this thesis demonstrates its flexibility and general applicability to a range of community-level biodiversity modeling scenarios. This comparative framework also enables a systematic assessment of how different modeling classes (statistical or machine learning) perform under varying

[2]Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 26 April 2026).

analytical conditions, thereby providing insight into their relative strengths, limitations, and appropriate use cases. The key research questions that this master's thesis aims to address are:

1. How do models perform with different strategies of cross-validation, specifically random cross-validation vs. spatial cross-validation?
2. How do models perform in data-limited scenarios?
3. How do different model classes (statistical vs. machine learning) perform compared to each other?

The results are therefore intended to illustrate the flexibility and practical utility of the pipeline, while demonstrating how standardized workflows can support more efficient and reproducible evaluation of top-down biodiversity models.

The remainder of this thesis is structured as follows. Section 2 provides background on biodiversity, including how it is defined and measured in Section 2.1, and reviews the main modeling approaches used to predict biodiversity patterns and trends in Section **??**. Section 3 describes the methodological framework, including the pipeline design in Section 3.1, the case study dataset in Section 3.2, and the models implemented in Section 3.3. Section 4 presents the results of the case study and illustrates the flexibility of the proposed pipeline. Section 5 discusses the implications of these results for biodiversity modeling and conservation, while Section 5.2 addresses relevant ethical considerations. Section 6 outlines potential future directions and applications for research in this area. Finally, Section 7 concludes the thesis by summarizing the key findings and their implications for biodiversity modeling and conservation.

2 Background

This section introduces biodiversity and how it is measured in Section 2.1. Section **??** provides a mathematical framework for understanding top-down modeling approaches, so we can use them for predicting community-level biodiversity metrics. This background section provides the necessary context for understanding the methods and results presented in this thesis. It also highlights the differences between the modeling approaches as well as the different assumptions and limitations of these models.

2.1 Biodiversity and how to measure it

Biodiversity is a complex and multifaceted concept that encompasses the variety of life on Earth, including the diversity of species, genetic diversity within species, and the diversity of ecosystems (Stork 2009). Humanity has relied on biodiversity for survival for thousands of years as it provides us with food, materials, medicine, water security, and has a substantial impact on human health and well-being (Gascon et al. 2015; Naeem et al. 2016; Brauman et al. 2020; Kim et al. 2024). Large losses of biodiversity would be catastrophic for humanity and ecosystems as it would lead to the loss of ecosystem functions, the collapse of food webs, adversely affect human health, degrade medicine production and discovery, and more (Roe, Seddon, and Elliott 2018). The rate of biodiversity loss is unprecedented, and we are quickly moving towards a sixth mass extinction (Cowie, Bouchet, and Fontaine 2022). To combat biodiversity loss effectively, conservation and ecological rehabilitation efforts require accurate, robust models that can predict biodiversity patterns, identify priority areas, and assess potential conservation outcomes (Sinclair, White, and Newell 2010; Pollock et al. 2020).

To effectively predict biodiversity, it is crucial to have a clear understanding of how biodiversity data is collected and measured. Generally, biodiversity data is collected through field surveys, remote sensing, and citizen science initiatives and other monitoring schemes depending on what is being measured (Proença et al. 2017; Beery et al. 2021; Brun et al. 2024). These approaches or protocols can provide data on species occurrences, abundances, and distributions across different spatial scales (Proença et al. 2017). Common biodiversity datasets can be categorized by the type of species information they contain. Presence-only data record locations where a species has been observed, but do not provide information on confirmed absences. Presence-absence data record whether each species is present or absent at a site. Abundance data provide counts or estimates of the number of individuals of each species at a site. Once these raw data is collected, there are three broad categories of biodiversity metrics that are computed: alpha diversity, beta diversity, and gamma diversity. Alpha diversity refers to diversity within a site, such as the diversity within a specific community or ecosystem. Beta diversity refers to the diversity between different communities or ecosystems. Gamma diversity refers to the species diversity across communities or ecosystems within a larger region (Andermann et al. 2022). For the purposes of this thesis, we will focus on alpha diversity.

Common alpha diversity metrics include species richness, absolute abundance, relative abundance, Shannon index, arithmetic mean abundance, and geometric mean abundance (Santini et al. 2017; Beery et al. 2021). Species richness is the most commonly used metric and simply counts the number of species in a given area (Ferrier and Guisan 2006). Species richness is a very popular metric because presence-absence data is the most common type of biodiversity data (Özkan Tümer et al. 2026). Absolute abundance measures the total number of individuals of each species at a site, and relative abundance measures the proportion of individuals of each species relative to the total number of individuals in a site (Beery et al. 2021). The Shannon index is another commonly used metric and combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community (Beery et al. 2021). Arithmetic mean abundance is the average abundance of species in a given area, and geometric mean abundance is the exponential of the average of the log abundances of species in a given area (Santini et al. 2017).

For the purposes of this thesis, it is beneficial to express these terms mathematically. Biodiversity metrics are usually calculated at the site level. A site is a spatial sampling unit that is defined according to a sampling protocol. We define a site as r, which is a spatial location, and within a biodiversity dataset there are n sites which make up the region of interest $R = \{r_1, r_2, \ldots, r_n\}$. Across the entire region of interest R there is a set S of unique species that were observed, and s is a specific species, where $s \in S$.

Furthermore, let a_{s,r_i} be the abundance or count of individuals of species s at site r_i.

The alpha diversity metrics used in this thesis are therefore defined at the site level as follows.

Species richness. Following Beery et al. (Beery et al. 2021), species richness is the number of unique species s at a given site r_i and can be expressed as:

$$SR_{r_i} = \sum_{s \in S} \mathbf{1}(a_{s,r_i} > 0), \tag{1}$$

where $\mathbf{1}(\cdot)$ is the indicator function that equals 1 if the condition is true and 0 otherwise. For reference, regional species richness across R can be expressed as:

$$SR_R = |S| \,. \tag{2}$$

Absolute abundance. Absolute abundance is the total number of individuals across species s at a given site r_i and can be expressed as:

$$A_{r_i} = \sum_{s \in S} a_{s,r_i}, \tag{3}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i (Beery et al. 2021), and $a_{s,r_i} = 0$ if species s is not present at site r_i.

Relative abundance. Relative abundance is the proportion of individuals of each species s relative to the total number of individuals at a given site r_i and can be expressed as:

$$p_{s,r_i} = \frac{a_{s,r_i}}{A_{r_i}}, \tag{4}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i and A_{r_i} is the total abundance of all species at site r_i (Beery et al. 2021). If $a_{s,r_i} = 0$ then species s is not present at site r_i, and $p_{s,r_i} = 0$.

Shannon index. The Shannon index combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community. It can be expressed as:

$$H_{r_i} = -\sum_{s \in S} p_{s,r_i} \ln(p_{s,r_i}), \tag{5}$$

where $p_{s,r_i}$is the relative abundance of species s at site r_i. If $p_{s,r_i} = 0$, then that term is not include in the calculation. The interpretation of the Shannon index is that higher values of H_{r_i} indicate higher diversity (Dušek and Popelková 2017).

Arithmetic mean abundance. Arithmetic mean abundance is the average abundance of species present at site r_i. It can be expressed as:

$$M_{r_i} = \frac{1}{SR_{r_i}} \sum_{s \in S} a_{s,r_i}, \tag{6}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i (Santini et al. 2017). If $a_{s,r_i} = 0$ for any species s in site r_i, then that species is not included in the calculation of the arithmetic mean abundance for site r_i.

Geometric mean abundance. Geometric mean abundance is the exponential of the average of the log abundances of species at a given site r_i and can be expressed as:

$$G_{r_i} = \exp\left(\frac{1}{SR_{r_i}} \sum_{s \in S} \ln\left(a_{s,r_i}\right)\right), \tag{7}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i (Santini et al. 2017). If $a_{s,r_i} = 0$ for any species s at site r_i, then that species is not included in the calculation of the geometric mean abundance for site r_i.

These metrics each have their own benefits and limitations. Species richness is the most commonly used metric because presence-absence data is the most common type of biodiversity data and it is easy to calculate (Lamb et al. 2009; Özkan Tümer et al. 2026). However, species richness does not account for the abundance of species and can be biased towards rare species (Ferrier and Guisan 2006). Absolute abundance and relative abundance capture more information about the abundance of species in a given area, but they can be dominated by common species (Beery et al. 2021). The Shannon index combines species richness and evenness into a single metric and can provide a more comprehensive measure of biodiversity, but it can be difficult to interpret (Dušek and Popelková 2017). Arithmetic mean abundance and geometric mean abundance are useful for measuring changes in overall populations but can be inflated by very high population density species (Santini et al. 2017). It is important to look at a wide range of metrics when measuring biodiversity as each metric measures different aspects of biodiversity and responds differently to changes in biodiversity (Lamb et al. 2009; Santini et al. 2017).

2.2 Top-Down Modeling Problem Formulation

Modeling biodiversity has historical roots in the early 20th century with work on modeling individual species distributions (Grinnell 1904). Since then models have expanded to include ecological models based on expert knowledge (Beery et al. 2021), parametric statistical models (Guisan, Edwards, and T. Hastie 2002), and machine learning models (Özkan Tümer et al. 2026). Community-level biodiversity models can be broadly divided into three strategies: bottom-up or "predict first, assemble later", joint or "assemble and predict together", and top-down or "assemble first, predict later' (Ferrier and Guisan 2006; Pollock et al. 2020). These strategies differ in how they produce predictions of community-level biodiversity metrics, and one of the key differences between top-down strategies and the other two is that top-down models directly predict community-level metrics. Top-down models directly output species richness or abundance, while the other two strategies first model the distribution of individual or multiple species and then aggregate those predictions to estimate species richness or abundance (Pollock et al. 2020). These top-down modeling approaches have been applied to predicting alpha, beta, and gamma biodiversity metrics across various spatial scales (C. Zhang et al. 2019; Hoskins et al. 2020; Mondal and Bhat 2021; Andermann et al. 2022).

Top-down modeling approaches come in a variety of forms, including regression-based models (Mondal and Bhat 2021), generalized dissimilarity models, which are specific to predicting beta diversity (Ferrier, Manion, et al. 2007; C. Zhang et al. 2019; Hoskins et al. 2020), and machine learning models (Andermann et al. 2022). To capture the broad range of top-down modeling approaches, we will define a mathematical framework that captures the shared structures of these models. Building on the notation from Section 2.1 and the formulations of Beery et al. (Beery et al. 2021) we define the following notation for a top-down modeling approach.

A site is denoted as $r \in \mathcal{R}$, where $\mathcal{R}$ is the spatial region of the study. Let $Y(r)$ denote the biodiversity response variable at site r. A general biodiversity dataset is defined as

$$\mathcal{D}_r = \{(r_i, y_i)\}_{i=1}^{n},$$

where $r_i \in \mathcal{R}$ is a sampled site and y_i is the observed realization of $Y(r_i)$. The response variable depends on the community-level biodiversity metric being measured and described in Section 2.1. For example, species richness and abundance may be represented as non-negative counts, while diversity metrics may be represented as non-negative real-valued indices. We then extract a representation of the environment in the region $\mathcal{R}$ as

$$h : \mathcal{R} \to \mathcal{X} \subseteq \mathbb{R}^p.$$

Thus,

$$h(r_i) = x_i$$

where x_i is the vector of predictor variables for site r_i and $x_i \in \mathcal{X} \subseteq \mathbb{R}^p$, and p is the number of predictor variables. The predictor variables are usually gathered from various sources, such as remote sensing data, climate datasets, and topographic maps (Waltari et al. 2014; Beery et al. 2021). Therefore, the dataset can be represented as

$$\mathcal{D}_x = \{(x_i, y_i)\}_{i=1}^{n}.$$

We then define a model

$$f_\Theta : \mathcal{X} \to \mathbb{R},$$
$$\hat{y}_i = f_{\hat{\Theta}}(x_i) = f_{\hat{\Theta}}(h(r_i)),$$

with fitted parameters $\hat{\Theta}$, where $\hat{y}_i$ is the model-based approximation of the community-level biodiversity response $Y(r_i)$.

These models are training in a supervised learning manner, where the dataset can be divided into training and testing sets for modeling training and evaluation.

$$\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n_{\text{train}}}, \quad \mathcal{D}_{\text{test}} = \{(x_i, y_i)\}_{i=n_{\text{train}}+1}^{n}$$

where n_{train} is the number of training examples and n is the total number of examples in the dataset $\mathcal{D}_x$. The model parameters Θ are estimated by minimizing a loss function that quantifies the difference between the observed responses y_i and the predicted responses $\hat{y}_i$ for a given observation. This is expressed as

$$\mathcal{L}(y_i, f_\Theta(x_i))$$

where $\mathcal{L}$ is the loss function, and can be mean squared error for regression tasks or cross-entropy for classification tasks, among others (Friedman 2001). A general supervised training objective is then to find parameters $\hat{\Theta}$ that minimize the average loss across all training examples:

$$\hat{\Theta} = \arg\min_{\Theta} \frac{1}{n_{\text{train}}} \sum_{i=1}^{n_{\text{train}}} \mathcal{L}(y_i, f_\Theta(x_i)),$$

and then the model's performance is evaluated on the test set. Here, the loss function refers to the objective function used during model training and the different model classes that are described in Section 2.3 use this objective function in different ways to find a good model. Evaluation metrics are used to assess the performance of the model on the test set, and these metrics can be different from the loss function used during training and model evaluation will be covered in Section 3.3.

For this thesis, the pipeline that we develop encompasses this entire problem formulation. The pipeline allows for flexibility in the choice of dataset $\mathcal{D}_r$, the choice of predictor variables x_i, and the choice of model f_Θ. The pipeline also includes steps for data preprocessing, model training, and model evaluation. The next section reviews the different types of models that were used for top-down community-level biodiversity modeling in this thesis.

2.3 Top-Down Models

This subsection reviews the different types of models that were used to test the pipeline for top-down community-level biodiversity modeling. The models discussed here are a subset of the model classes that can be used for top-down community-level biodiversity modeling. We define the assumptions and limitations of these models. The models discussed include generalized linear models (Nelder and Wedderburn 1972), generalized additive models (T. J. Hastie and Tibshirani 1986), random forests (Breiman 2001b), gradient boosting (Friedman 2001), and neural networks (Bishop 2006). These models are used to demonstrate the effectiveness of the pipeline.

2.3.1 Generalized Linear Models

Generalized Linear Models (GLMs) are an extension of linear regression models that allow the response variable to follow a distribution from the exponential family (Nelder and Wedderburn 1972). Under normal linear regression, the response variable Y is modeled as

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon,$$

where ϵ is the error term. Linear regression assumes that the conditional mean of the response variable Y is linear in the predictors $x_1, x_2, \cdots, x_p$, the error term ϵ is normally distributed with mean 0 and variance σ^2 or $\epsilon \sim \mathcal{N}(0, \sigma^2)$, the error term ϵ is independent and identically distributed, and the variance of Y is constant across observations (Guisan, Edwards, and T. Hastie 2002).

In GLMs, the response variable for the i-th observation, Y_i, is assumed to follow a distribution from the exponential family, and a link function $g(\cdot)$ is used to connect the linear predictor to the mean of the response variable. The model is then specified as

$$g(\mathbb{E}(Y_i|\mathbf{x}_i)) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip},$$

where $\mathbb{E}(Y_i|\mathbf{x}_i) = \mu_i$ is the expected value of Y_i given the predictors $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and the linear predictor is $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ (Guisan, Edwards, and T. Hastie 2002). Then, we can rewrite the model as

$$g(\mu_i) = \eta_i.$$

To make predictions with a GLM, we use the inverse of the link function $g^{-1}(\cdot)$ to transform the linear predictor back to the scale of the response variable. The predicted value of the expected value for the i-th observation is then

$$\hat{\mu}_i = g^{-1}(\hat{\eta}_i) = g^{-1}(\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}).$$

Common link functions include the identity link for Gaussian response variables (Guisan, Edwards, and T. Hastie 2002) and a log link function for count data (Cai et al. 2023). Fitting this model is done by maximizing the likelihood of the observed data under the model, which can be done through iterative weighted least squares (Nelder and Wedderburn 1972; Guisan, Edwards, and T. Hastie 2002).

The main advantage of a GLM is that the response variable does not need to be normally distributed. Instead, GLMs allow the response to follow a distribution from the exponential family, such as Poisson models for count data and negative binomial models for overdispersed counts (Nelder and Wedderburn 1972; Cai et al. 2023), which provides greater flexibility for ecological data (Guisan, Edwards, and T. Hastie 2002). This is particularly useful for biodiversity metrics that are non-normal or overdispersed. For example, species richness is count data and often exhibits higher variance than expected under a Poisson model, making the negative binomial distribution a more appropriate choice (Cai et al. 2023). Another benefit of GLMs is that they are interpretable as the coefficients β_j represent the change in the linear predictor, or the transformed expected response $g(\mu_i)$, associated with a one-unit change in the predictor variable x_{ij} while holding all other predictors constant, and the interpretation depends on the choice of link function.

GLMs have been used as a top-down modeling strategy for predicting community-level biodiversity metrics (Brugere et al. 2023; Cai et al. 2023; Ngor et al. 2023). While GLMs are more flexible than linear regression models and have been shown to be effective for single species distribution modeling (Norberg et al. 2019), they can have limited performance when predicting community-level biodiversity metrics such as species richness (Robin and Hasif 2021; Ngor et al. 2023; Cai et al. 2023). First, GLM performance can be strongly influenced by the choice of predictor variables (Guisan, Edwards, and T. Hastie 2002). Second, GLMs may be insufficiently flexible when the true relationship between predictors and the response is highly nonlinear or complex (Yee and Mitchell 1991). Third, GLMs require the analyst to specify an appropriate response distribution, link function, and model structure, which should be chosen carefully to match the ecological context (Austin 2007). Although the latter limitations are often discussed in the context of species distribution modeling, they are also relevant to top-down models of community-level biodiversity metrics.

2.3.2 Generalized Additive Models

Generalized Additive Models (GAMs) are an extension of GLMs that allow for non-linear relationships between the predictors and the response variable (T. J. Hastie and Tibshirani 1986). In a GAM, the linear predictor is generalized to an additive predictor composed of smooth functions of the covariates. The model can be specified as

$$g(\mu_i) = \eta_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}),$$

where $\mu_i = \mathbb{E}(Y_i \mid \mathbf{x}_i)$, $g(\cdot)$ is the link function, and $f_j(\cdot)$ is a smooth function of the predictor variable x_{ij} (T. J. Hastie and Tibshirani 1986). The use of smooth functions in GAMs provides greater flexibility than GLMs, as they can capture complex, non-linear relationships between the predictors and the transformed expected response $g(\mu_i)$. The smooth functions themselves are very flexible and can be represented by many functions, such as splines, but are also commonly estimated from the data itself (T. J. Hastie and Tibshirani 1986; Yee and Mitchell 1991). The link function $g(\cdot)$ still must connect the expected response μ_i to the additive linear predictor η_i (T. J. Hastie and Tibshirani 1986). GAMs are also fitting in a similar manner to GLMs and can be fitting through local likelihood estimation (T. J. Hastie and Tibshirani 1986).

The increased flexibility of GAMs has also contributed to their success in modeling individual species distributions (Valavi et al. 2022), and they have been utilized to predict community-level biodiversity metrics such as species richness and have shown better predictive performance than GLMs (Mondal and Bhat 2021; Cai et al. 2023). GAMs are also interpretable, as each smooth function f_j can be plotted as a function of its corresponding predictor variable x_{ij}, showing how the expected response changes across values of the predictor. However, some of the limitations of GAMs are similar to those of GLMs. First, GAM performance can also be strongly influenced by the choice of predictor variables (Guisan, Edwards, and T. Hastie 2002). While GAMs are more flexible than GLMs, this also increases model complexity and can increase computational cost (Mondal and Bhat 2021).

2.3.3 Random Forests

Random Forests are a supervised machine learning model that is based on the idea of combining several decision trees in an ensemble method to improve predictive performance (Breiman 2001b). To understand what a Random Forest is more concretely, we will first describe a decision tree. Given a set of predictor variables $X_1, X_2, \cdots, X_p$ and a response variable Y, a decision tree will recursively partition the predictor space into k regions such that the response variable is as homogeneous as possible within each region (Loh 2011). For a classification task with k classes, training a decision tree involves splitting the predictor space into regions $R_1, R_2, \cdots, R_m$ such that in each region R_j the predicted class is the majority class of the training observations that fall into that region (Moisen 2008; Loh 2011). For a regression task, the predicted value for each region R_j is the average of the response variable for the training observations that fall into that region (Moisen 2008; Loh 2011). The process of splitting the predictor space is done by searching over all predictor variables and candidate threshold values for each predictor. The best split is based on a criterion or loss function such as minimizing the variance or the mean squared error (MSE) (Klusowski 2020). For a regression tree, first, we would look at the sample variance or MSE of the node t, before the split, which is calculated as

$$MSE_t = \frac{1}{n_t}\sum_{i=1}^{n_t}(y_i - \bar{y}_t)^2,$$

where n_t is the number of observations, y_i is the response variable for the i-th observation in node t, and $\bar{y}_t$ is the mean among the observations in node t. Then, for a given split that divides the node t into two child nodes t_L and t_R with n_L and n_R observations, respectively, we would calculate the change in MSE. This can be expressed as

$$\Delta(MSE) = MSE_t - (\frac{n_L}{n_t}MSE_{t_L} + \frac{n_R}{n_t}MSE_{t_R}).$$

The MSE for the left and right children is calculated in the same manner as MSE_t but using the observations in the left and right children, respectively. The best split is the one that maximizes $\Delta(MSE)$ or equivalently minimizes the MSE of the two sub-samples proportionally (Klusowski 2020). This calculation is repeated for several potential splits across all predictor variables, and the best split is chosen based on this criterion. This process is repeated recursively on every subset until a stopping condition is met, such as a minimum number of observations in a leaf node or a maximum tree depth (Mienye and Jere 2024). An illustration of a decision tree can be seen in Figure 1.

A Random Forest is an ensemble method that combines hundreds or thousands of decision trees to produce a prediction. Each tree is trained on a bootstrapped sample of the data, while each split considers only a random subset of the predictor variables. For regression tasks, the final prediction is obtained by averaging the predictions across all trees; for classification tasks, it is typically determined by majority vote (Breiman 2001b; Mienye and Jere 2024). This bootstrapping and aggregation, commonly referred to as bagging, also helps reduce the variance of the final model (Aria, Cuccurullo, and Gnasso 2021).

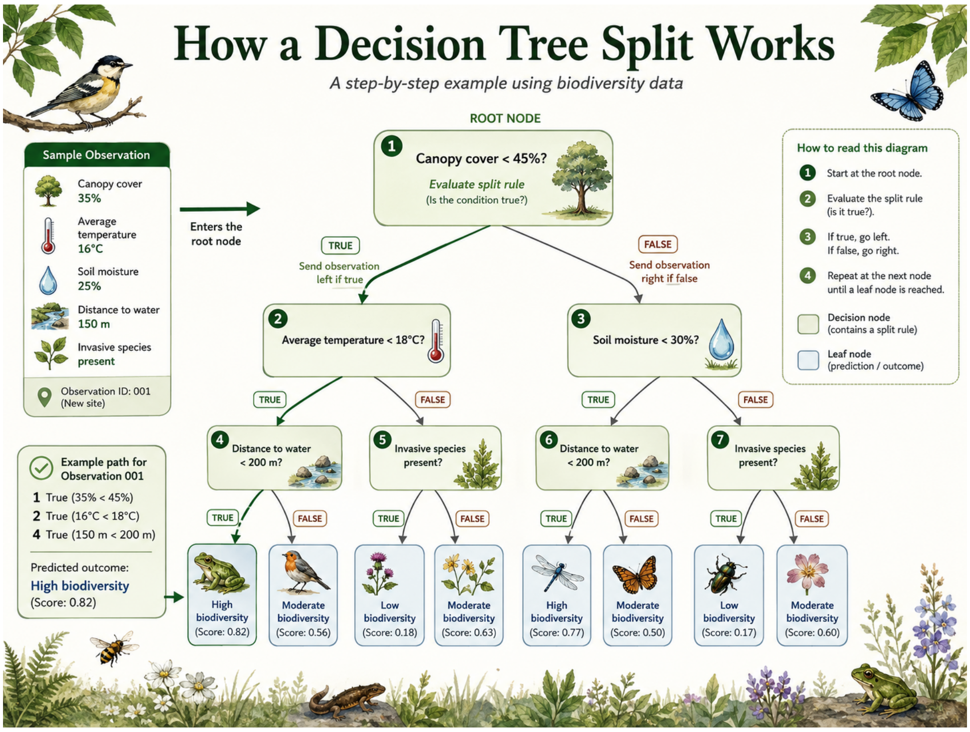

Figure 1: An illustrative decision tree for a regression task. The tree is built by recursively splitting the predictor space. Image generated with ChatGPT using the prompt: "Generate an image of a decision tree that shows how a split is processed. Use a biodiversity theme for the variables."

Random Forests are a popular machine learning model for top-down community-level biodiversity modeling (Olaya-Marín, Martínez-Capel, and Vezza 2013; C. Zhang et al. 2019; Večeřa et al. 2019; Ngor et al. 2023; Cai et al. 2023; Brugere et al. 2023; Baggström et al. 2025) due to their ability to capture complex, non-linear relationships between predictors and the response variable, and their robustness to overfitting due to their ensemble nature (Breiman 2001b). However, some of their shortcomings include being less interpretable than GLMs and GAMs because they are an ensemble of many, possibly deep, trees (Aria, Cuccurullo, and Gnasso 2021). Also, random forests can still show a gap between training and test performance (Olaya-Marín, Martínez-Capel, and Vezza 2013; Baggström et al. 2025). Furthermore, they can become computationally expensive when the number of trees is large or when the depth of the trees is large.

2.3.4 Gradient Boosting

Gradient boosting is another ensemble method but is built in a sequential manner instead of in parallel like Random Forests. The idea is to fit a sequence of weak learners, typically small decision trees, where each subsequent tree is trained to correct for the errors of the previous trees (Friedman 2001; Lindholm et al. 2022). The final model is an additive model of the form

$$f_M(x) = f_0(x) + \sum_{m=1}^{M} \beta_m h_m(x),$$

where $f_0(x)$ is an initial guess, h_m is the m-th weak learner, and β_m is the weight for that learner (Friedman 2001). The model is trained in an iterative manner, where each iteration fits a new weak learner to the negative gradient of the loss function with respect to the current model's predictions (Friedman 2001). Mathematically, this is expressed as

$$g_{im}(x) = -\left[\frac{\partial \mathcal{L}(y_i, f(x_i))}{\partial f(x_i)}\right]_{f=f_{m-1}},$$

where g_{im} then becomes the response variable for the m-th weak learner h_m (Friedman 2001). Another way to think of it is that the dataset for h_m is $\mathcal{D} = \{(x_i, g_{im}(x_i))\}_{i=1}^n$ (Lindholm et al. 2022). The weak learner h_m is then trained to fit these negative gradients. Once h_m is trained, β_m is then used to weight the contribution of the weak learner h_m, and is calculated as:

$$\beta_m = \arg\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(y_i, f_{m-1}(x_i) + \beta_m h_m(x_i)),$$

where n is the number of observations, and β_m is usually found through some sort of minimization optimization such as line search (Friedman 2001). If the loss function is the squared error, then the negative gradient is simply the residuals of the current model's predictions, and the weak learner is trained to fit these residuals. The general algorithm can be summarized as follows:

Algorithm 1 Gradient Boosting Algorithm

Require: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, number of weak learners M, learning rate ν
1: Initialize the model with an initial guess: $f_0(x)$
2: **for** $m = 1$ to M **do**
3: Compute the negative gradients: $g_{im}(x) = -\left[\frac{\partial \mathcal{L}(y_i, f(x_i))}{\partial f(x_i)}\right]_{f=f_{m-1}}$
4: Fit a weak learner h_m to the negative gradient: $h_m = \arg\min_h \sum_{i=1}^n \mathcal{L}(g_{im}(x_i), h_m(x_i))$
5: Compute the weight for the weak learner: $\beta_m = \arg\min_\beta \frac{1}{n} \sum_{i=1}^n \mathcal{L}(y_i, f_{m-1}(x_i) + \beta h_m(x_i))$
6: Update the model: $f_m(x) = f_{m-1}(x) + \nu \beta_m h_m(x)$
7: **end for**

An extension of gradient boosting that is relevant to community-level biodiversity modeling is categorical boosting, or CatBoost. The main improvements to normal gradient boosting are that CatBoost natively handles categorical features, uses ordered boosting to deal with conditional shift, and uses oblivious trees as the weak learners. CatBoost natively handles categorical features by computing a target statistic for each category by computing it from the previous histories. Ordered boosting deals with conditional shift, which is where the distribution between training and test data differ, by permuting the data and training model M_i only on the data that comes before the i-th observation in the permutation. This is not feasible in practice and instead is done through approximation. Oblivious trees are a special type of decision tree where the same splitting criterion is used at every level of the tree, which allows for balanced trees that are less prone to overfitting (Prokhorenkova et al. 2018).

Gradient boosting methods are also popular for community-level biodiversity modeling (Cai et al. 2023; Baggström et al. 2025) and also for modeling the value of the ecosystem (Liu et al. 2024). However, some limitations of gradient boosting are the conditional shift problem. This is when the distribution of the training data differs from the distribution of the test data (K. Zhang et al. n.d.; Prokhorenkova et al. 2018). Gradient boosting can also overfit if the learning rate isn't properly tuned (Friedman 2001). Additionally, gradient boosting can be computationally expensive, especially when the number of weak learners M is large or when the weak learners are complex.

2.3.5 Neural Networks

Neural Networks are a flexible class of machine learning models that are used widely for many applications. Here we will discuss artificial neural networks (ANNs) and specifically Feed Forward Neural Networks (FFNNs), which are commonly used in community-level biodiversity modeling (Andermann et al. 2022; Brugere et al. 2023; Cai et al. 2023). Neural networks are composed of nodes that take in input data, apply a linear transformation, and then apply a non-linear activation function to produce an output. The nodes are organized into layers, and the output of one layer serves as the input to the next layer (Bishop 2006; Lindholm et al. 2022). Mathematically, this can be expressed as

$$\begin{aligned} \mathbf{h}_0 &= \mathbf{x}, \\ \mathbf{h}_l &= \sigma_l(\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l), \quad l = 1, 2, \ldots, L, \\ f_\Theta(\mathbf{x}) &= \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1}. \end{aligned}$$

where $\mathbf{x}$ is the input vector, $\mathbf{W}$ is the weight matrix, $\mathbf{b}$ is the bias vector, σ is the activation function, and $\mathbf{h}$ is the output of the layer. The dimensions are as follows:

$$\mathbf{x} \in \mathbb{R}^p, \qquad \mathbf{W}_l \in \mathbb{R}^{q_l \times q_{l-1}}, \qquad \mathbf{b}_l \in \mathbb{R}^{q_l}, \qquad \mathbf{h}_l \in \mathbb{R}^{q_l},$$

where p is the number of input features, $q_0 = p$, and q_l is the hidden size of layer l. The activation function σ is a non-linear function, and common choices include the rectified linear unit (ReLU), $\sigma(x) = \max(0, x)$, and the sigmoid function, $\sigma(x) = \frac{1}{1+e^{-x}}$. The size of the output layer depends on the type of task, for regression it would have one node, and for classification it would have k nodes that are usually standardized according to the softmax function, which converts the outputs to be probabilities that sum to one (Lindholm et al. 2022; Prince 2023).

The parameters of the neural network are the weights and biases $\Theta = \{\mathbf{W}_l, \mathbf{b}_l\}_{l=1}^{L+1}$. There is also a loss function $\mathcal{L}(\mathbf{y}, f_\Theta(\mathbf{x}))$ that quantifies the difference between the observed responses $\mathbf{y}$ and the predicted responses $f_\Theta(\mathbf{x})$. The parameters of the neural network are learned during training by minimizing this loss function through the backpropagation algorithm(Lindholm et al. 2022). In backpropagation first, we will evaluate the loss of the network $\mathcal{L}_t(\mathbf{y}_b, f_{\Theta_t}(\mathbf{X}_b))$ at time t with a mini-batch of data $\mathbf{X}_b \subset \mathbf{X}$, where $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the full input matrix and $\mathbf{y}_b$ is the corresponding observed responses to the mini-batch $\mathbf{X}_b$. Then we will update the network parameters Θ by taking a step in the direction of the negative gradient of the loss function with respect to the parameters. This can be expressed as

$$\Theta_{t+1} = \Theta_t - \alpha \nabla_\Theta \mathcal{L}_t(\mathbf{y}_b, f_{\Theta_t}(\mathbf{X}_b)),$$

where α is the learning rate (Lindholm et al. 2022; Prince 2023). This will be repeated for several iterations, and each mini-batch will be a different subset of the training data without replacement when all mini-batches have been used, which is called one epoch. The training process will be repeated for several epochs, where new mini-batches are drawn each epoch, until the model converges or a stopping criterion is met, the updating of these parameters is often called stochastic gradient descent (Lindholm et al. 2022).

The gradients of all the parameters are calculated through the backpropagation algorithm, which uses the chain rule of calculus to compute the gradients efficiently. For example, we can calculate the gradients of layer l by using the partial derivatives of the layers above it. If we denote $\mathbf{z}_l = \mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l$ as the pre-activation value and the activation value $\mathbf{h}_l = \sigma_l(\mathbf{z}_l)$, then the gradient of the loss function with respect to the parameters of the final layer $L + 1$ can be written as

$$\begin{aligned} \frac{\partial \mathcal{L}_t}{\partial \mathbf{W}_{L+1}} &= \frac{\partial \mathcal{L}_t}{\partial \mathbf{z}_{L+1}} \frac{\partial \mathbf{z}_{L+1}}{\partial \mathbf{W}_{L+1}} \\ \frac{\partial \mathcal{L}_t}{\partial \mathbf{b}_{L+1}} &= \frac{\partial \mathcal{L}_t}{\partial \mathbf{z}_{L+1}} \frac{\partial \mathbf{z}_{L+1}}{\partial \mathbf{b}_{L+1}}. \end{aligned}$$

Concurrently, the gradient of the loss function with respect to a general output of layer l and $\mathbf{W}_l$ and $\mathbf{b}_l$ can be written as

$$\begin{aligned} \frac{\partial \mathcal{L}_t}{\partial \mathbf{z}_l} &= \frac{\partial \mathcal{L}_t}{\partial \mathbf{z}_{l+1}} \frac{\partial \mathbf{z}_{l+1}}{\partial \mathbf{h}_l} \frac{\partial \mathbf{h}_l}{\partial \mathbf{z}_l} \\ \frac{\partial \mathcal{L}_t}{\partial \mathbf{W}_l} &= \frac{\partial \mathcal{L}_t}{\partial \mathbf{z}_{l+1}} \frac{\partial \mathbf{z}_{l+1}}{\partial \mathbf{h}_l} \frac{\partial \mathbf{h}_l}{\partial \mathbf{z}_l} \frac{\partial \mathbf{z}_l}{\partial \mathbf{W}_l} \\ \frac{\partial \mathcal{L}_t}{\partial \mathbf{b}_l} &= \frac{\partial \mathcal{L}_t}{\partial \mathbf{z}_{l+1}} \frac{\partial \mathbf{z}_{l+1}}{\partial \mathbf{h}_l} \frac{\partial \mathbf{h}_l}{\partial \mathbf{z}_l} \frac{\partial \mathbf{z}_l}{\partial \mathbf{b}_l}, \end{aligned}$$

for $l = L, L-1, \ldots, 1$. We can continue to expand these partial derivatives until we reach the input layer, which allows us to compute the gradients for all parameters in the network (Prince 2023).

Neural Networks are a very powerful and flexible class of models that can capture complex, non-linear relationships between predictors and the response variable. They have been used for top-down community-level biodiversity modeling (Mondal and Bhat 2021; Andermann et al. 2022; Brugere et al. 2023; Cai et al. 2023; Hu, Si-Moussi, and Thuiller 2025) and have shown good predictive performance. However, Neural Networks are widely considered as "black boxes" as it can be very difficult to interpret these models because they are so complex (Pichler and Hartig 2023; Hu, Si-Moussi, and Thuiller 2025). Another limitation of neural networks is that they frequently have large data requirements and can take a long time to train, especially when the network architecture is large or when the dataset is large (Pichler and

Hartig 2023). Neural Networks are also cannt extrapolate well outside the range of the training data, which can be a problem for community-level biodiversity modeling when we want to make predictions in new areas or under future climate scenarios (Beery et al. 2021; Pichler and Hartig 2023).

2.4 The current top-down modeling landscape

Community-level biodiversity modeling can be computationally and methodologically intensive because study areas can span spatial scales ranging from countries (Baggström et al. 2025), to continents (Davis et al. 2023), to the entire globe (Hoskins et al. 2020). Furthermore, the number of species used for community-level biodiversity metrics can be in the hundreds (Andermann et al. 2022) to hundreds of thousands (Hoskins et al. 2020), and selection of predictor variables is a non-trivial task (Williams et al. 2012). Community-level biodiversity models have been proposed as an important direction to explore with further research (Ferrier, Drielsma, et al. 2002; Ferrier and Guisan 2006), and the potential for advancing these models, including top-down approaches, remains underdeveloped along with rigorous testing of these models for ecological inference and prediction (Pollock et al. 2020). While there are studies demonstrating the potential of top-down approaches each study has to employ their own workflow which can become cumbersome and makes comparison difficult. Therefore, the development of a pipeline that allows for quick exploration of different biodiversity datasets, predictor variables, and community-level biodiversity models is an important contribution to this field.

3 Methods

In this section we describe the methods used to model biodiversity at the community level. We first outline our major contribution—a pipeline for spatial biodiversity modeling—and then describe the data, model selection process, and evaluation metrics used in our study.

3.1 Spatial Biodiversity Modeling Pipeline

The spatial biodiversity modeling pipeline, hereafter referred to as the SBM pipeline, we developed is based on the work of Nyström (Nyström 2024), and consists of the following three main components: data processing, feature engineering, and model training. We designed the SBM-pipe to be flexible in terms of the types of data and models that can be used, allowing for a wide range of applications in biodiversity modeling. The goal of the SBM pipeline is to facilitate the development and evaluation of top-down community-level biodiversity models, and to provide a tool for ecologists and researchers to explore the potential of these models in predicting biodiversity patterns and trends. The pipeline is implemented in Python along with a configuration file and for each component is available on GitHub (Caulk and Nyström 2026), allowing for easy access and collaboration among researchers in the field of biodiversity modeling. A diagram of the pipeline is shown in Figure 2, and a more detailed description of each step is provided in the following sections.

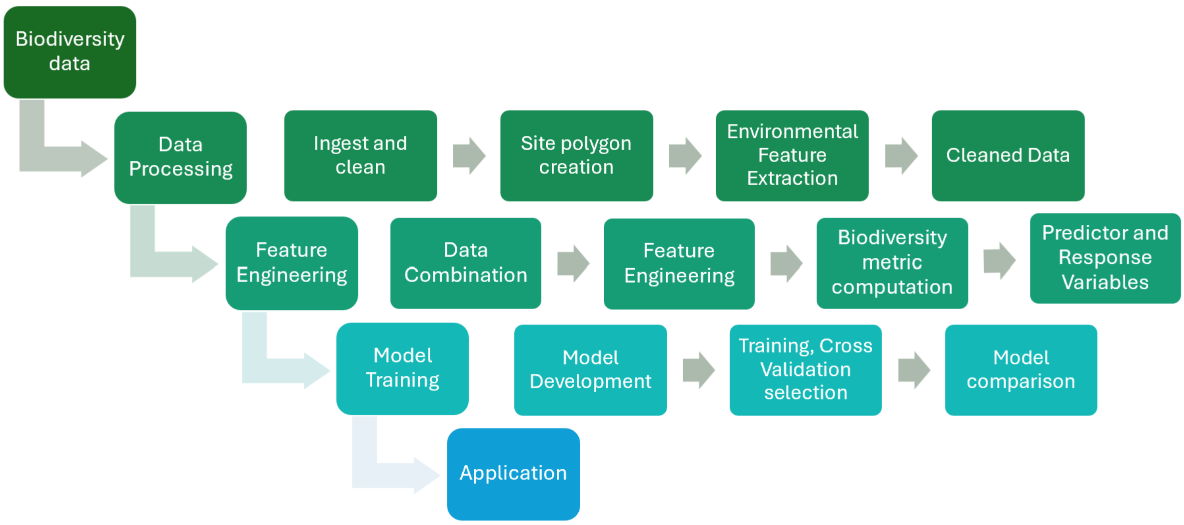

Figure 2: Overview of the Spatial Biodiversity Modeling Pipeline (SBM-pipe). The pipeline has 3 main components: data processing, feature engineering, and model training. Each component consists of several steps that are designed to be flexible and adaptable to different types of biodiversity data and modeling approaches. The pipeline is implemented in Python and is available on GitHub (Caulk and Nyström 2026).

3.1.1 Biodiversity data input

The first step in the SBM pipeline is to select appropriate biodiversity data. Researchers can draw on a wide range of biodiversity data sources, including citizen-science initiatives, government-funded monitoring programmes, and researcher-collected datasets (Zipkin et al. 2023); however, to standardize the data input we chose to use the Darwin Core (DwC) format (Wieczorek et al. 2012). This format was chosen because it is a flexible and straightforward format for biodiversity data. One of the largest biodiversity data repositories, GBIF, standardizes its data according to this format and the data used specifically to test the SBM pipeline is in Section 3.2. While this format is widely used, it is important to note that there are many other formats for biodiversity data, and the SBM pipeline is designed to account for this by allowing other formats as long as the user provides a mapping to DwC format.

3.1.2 Data Processing

Once the biodiversity dataset is chosen by the user, the next step in the SBM pipeline is to process the data. This involves several steps, including ingesting and cleaning the data, site polygon creation, and environmental feature extraction. In the ingestion and cleaning step the data is read into the pipeline and any necessary cleaning steps are performed. This includes removing any records with missing or invalid data, removing rows that don't have precise location information, removing records that are outside the dataset's date range, removing exact duplicates, validating location identification integrity, and inferring organism quantity [3]. Important steps here involve removing rows that are not precise enough for spatial modeling, this includes records whose latitude and longitude values are not precise enough, usually less than four decimal places, and the precision of these values can be specified by the user. Another important step is validating location identification integrity, if a biodiversity dataset is conduct over many decades, it is possible that the same unique location identifier (location ID) may change. This results in many location IDs mapping to the same latitude and longitude values, which can cause issues for spatial modeling. To address this, the pipeline checks for duplicate location IDs on latitude and longitude values and will rename location IDs to ensure that each unique location ID corresponds to a unique latitude and longitude pair. Finally, the pipeline infers organism quantity by checking for the presence of the "individualCount", "organismQuantity", and "organismQuantityType" fields in the dataset which are apart of DwC format (Wieczorek et al. 2012). If these fields are not present then the pipeline will infer organism quantity by counting the number of records for each unique location ID, each unique species, and each unique date as sometimes abundance data is not always provided in the default fields. If that data is truly occurrence data (presence/absence) this will have no effect on the dataset. All of these steps can be specified by the user in the configuration file, allowing for flexibility in the data processing step.

After the data is ingested and cleaned, the next step is to create site polygons. This involves creating polygons around each unique latitude and longitude pair in the dataset at different scales. This is done because biodiversity datasets usually vary in spatial scale (Sicacha-Parada et al. 2022). The polygons are also created so we can later extract environmental features from these polygons. The user can define the radius for the site polygons as well as the shape of these polygons. If the dataset contains coordinate uncertainty information, then the pipeline will also create polygons with a radius equal to the coordinate uncertainty value. These polygons are projected both in a global coordinate reference system (CRS) and a local CRS, which is determined by the latitude and longitude values of each site. The global CRS, EPSG:4326, is used for extracting environmental features, while the local CRS is used for visualization.

Once every site polygon is created, the next step is to extract environmental features from these polygons. The environmental features are extracted from datasets provided by the user as the choice of environmental features can vary and is not the same for every problem (Williams et al. 2012). The environmental feature datasets should be provided in a raster format, this will allow the SBM pipeline to extract features from the site polygons. The user can also specify the type of summary statistic to use when extracting features from the site polygons, this includes mean, median, min, max, and standard deviation.

3.1.3 Feature Engineering

Once the data has been processed, site polygons created, and environmental features extracted, the next step in the SBM pipeline is to perform feature engineering. The first step in feature engineering is to combine the biodiversity data with the extracted environmental features. This involves joining the biodiversity data with the environmental features based on the unique location ID for each site and time dimension which is usually the year. This results in a dataset that contains both the biodiversity data and the environmental features for each site and time dimension.

Once we have our combined biodiversity and environmental feature dataset, the next step is feature engineering. Feature engineering involves creating new features from the existing data that may be more informative for modeling. This can include creating interaction terms between environmental features,

[3] GBIF data quality guidance: https://docs.gbif.org/course-data-mobilization/en/data-quality.html (accessed 2026-05-12).

creating polynomial features, or other advanced feature engineering techniques (Katya 2023). Currently, the feature engineering in the SBM pipeline will log, square, and square root transform the continuous environmental features.

An important step of the feature engineering process is to calculate the community-level biodiversity metrics that will be used as the target variable for modeling. This includes calculating alpha diversity, which is a measure of the diversity within a single site. Currently the SBM pipeline calculates all the alpha diversity metrics described in Section 2.1. This includes species richness, absolute abundance, relative abundance, Shannon index, arithmetic mean abundance, and geometric mean abundance. Furthermore, when calculating these alpha diversity metrics, the user can define the taxonomic resolutions to group at before calculating these metrics. By default no taxonomic grouping done is always included. This type of taxonomic grouping is sometimes used when species-level identification is difficult, costly, or inconsistent (De Oliveira et al. 2020).

At the end of the feature engineering step, we have a datasets that contains community-level biodiversity metrics as well as the environmental features for each site and time dimension and at varying taxonomic resolutions. These datasets can then be used for modeling in the next step of the SBM pipeline.

3.2 Data

3.3 Model Selection

3.4 Model Evaluation Metrics

4 Results/ Case Study

5 Discussion

5.1 Findings

We designed the pipeline to be flexible and adaptable to different types of biodiversity data and modeling approaches.

5.2 Ethical Discussion

6 Future Work/ Applications

Future Work includes a full comprehensive benchmark of community level models across a wide range of datasets and models. The integration with GBIF should should allow for easy facilitation of this. This would allow for a more systematic evaluation of community level models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a comprehensive benchmark would allow for a more nuanced evaluation of community-level models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

Talk about comparisons of top-down methods with bottom-up or assemble and predict together as future works. There are some in the Zotero on SDM vs. BDMs folder.

7 Conclusion

A Appendix A: Github Repository

Includes the instructions there

B Appendix B: Additional Results

Includes the additional results there

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